

The Banach-Butterfly Invariant: Influence-Adaptive Walsh Geometry for Ternary Polynomial Threshold Functions

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Abstract

We introduce the *Banach-Butterfly Transform* (BBT), a spectral framework that equips each layer of the Walsh-Hadamard butterfly factorization with an influence-adaptive ℓ_p geometry. For a Boolean function $f : \{-1, +1\}^n \rightarrow \{-1, +1\}$ with coordinate influences $\text{Inf}_\ell(f)$, the BBT assigns exponent $p_\ell = 1 + \text{Inf}_\ell(f) \in [1, 2]$ to butterfly layer ℓ , yielding function-dependent operator norms $\|A^{-1}\|_{p_\ell \rightarrow p_\ell} = 2^{-\text{Inf}_\ell/(1+\text{Inf}_\ell)}$ and a computable *contraction invariant* $\mu(f) = \prod_\ell 2^{-\text{Inf}_\ell/(1+\text{Inf}_\ell)}$.

We prove that $\log_2 \mu(f)$ satisfies the tight bound $\log_2 \mu(f) \geq -I(f)/(1+I(f)/n)$ via Jensen’s inequality applied to the concave function $\varphi(x) = x/(1+x)$, and that μ is *strictly Schur-convex* in the influence vector (more concentrated influence $\Rightarrow$ strictly larger μ , modulo permutation), yielding distinct scaling classes: $\mu \sim 2^{-n/2}$ for parity (smallest μ among the canonical examples), $\mu \sim 2^{-\Theta(\sqrt{n})}$ for majority, and $\mu = 2^{-1/2}$ for dictators (constant in n). We show that $\log_2 \mu(f)$ is a *rational* (not polynomial) function of Fourier coefficients while $\mu(f)$ is *algebraic*, and that μ separates functions indistinguishable by total influence alone—122 such pairs exist at $n = 3$.

Using the certified $n \leq 4$ ternary Walsh-threshold universe from the companion synthesis manuscript [1] as a finite controlled testbed, we evaluate BBT diagnostics on all 65,536 Boolean functions at $n = 4$. Specifically, we compute exact minimum-support MILP certificates for every function and regress minimum support against μ , I , influence entropy, and a layerwise cancellation index. We find that μ shows a strong conditional correlation with minimum support within the largest fixed-influence stratum at $n = 4$ (Spearman $\rho = +0.571$); under both function-uniform and NPN-canonical sampling at $n = 5$, this relationship reverses sign ($\rho \approx -0.38$), indicating that μ is a valid Schur-convex concentration invariant but not a universal monotone predictor of minimum support across all n .

A companion application paper [2] validates a real-valued spectral-energy proxy inspired by this theory on five pretrained LLMs at W2A16 (group size = 64), reducing wikitext-2 perplexity by 15–58% relative to vanilla **auto-round**. We summarise that empirical line of work in Section 8.5 and refer the reader to the companion paper for the full results; the real-valued WHT activation-energy quantity used there is *not* the Boolean influence of this paper, and the connection to the Schur-convexity theorem is qualitative, not formal.

Relation to companion work. A companion manuscript currently under review [1] establishes (and we cite) the certified ternary representability of every Boolean function on at most four variables, via differentiable spectral coefficient selection plus Sinkhorn-constrained composition. The present paper does *not* re-prove that universality result — it reuses the certified $n \leq 4$ universe as a finite controlled testbed and asks a different question: *which functions are geometrically harder to represent in the Walsh basis, and can intrinsic Fourier-analytic quantities*

predict that difficulty? The new contributions here are the influence-adaptive Banach butterfly profile, its Schur-convexity / majorization theory, its rational-vs-polynomial separation properties, and its empirical correlation with support, margin, and cancellation diagnostics. The certified $n \leq 4$ result is shared, but it is not the headline of this paper.

1 Introduction

A fundamental question in the analysis of Boolean functions is: given $f : \{-1, +1\}^n \rightarrow \{-1, +1\}$, when can f be represented as the sign of a sparse linear combination of Walsh characters? Concretely, we ask for the existence of a *ternary polynomial threshold function* (PTF):

$$f(x) = \text{sign} \left(\sum_{S \subseteq [n]} w_S \chi_S(x) \right), \quad w_S \in \{-1, 0, +1\}, \quad (1)$$

where $\chi_S(x) = \prod_{i \in S} x_i$ are the Walsh characters forming an orthonormal basis under the uniform distribution on $\{-1, +1\}^n$.

The appeal of ternary PTFs is both theoretical and practical. Theoretically, their existence connects to spectral concentration [5], polynomial threshold complexity [4], and the structure of the Fourier spectrum. Practically, ternary weights $\{-1, 0, +1\}$ enable zero-cost quantization: inference reduces to additions and subtractions with no multiplication, achieving single-cycle combinational logic on hardware [15].

Standard spectral analysis of Boolean functions operates in $\ell_2(\mathbb{Z}_2^n)$, where the Walsh-Hadamard Transform (WHT) is unitary (up to scaling) and Parseval’s identity gives $\sum_S \hat{f}(S)^2 = 1$. This ℓ_2 structure is universal: the butterfly factorization of the WHT has eigenvalue $\sqrt{2}$ at every layer regardless of f . Consequently, ℓ_2 spectral analysis treats all Boolean functions identically at the structural level.

The BBT idea. We break this universality by equipping each butterfly layer with an *influence-adaptive* ℓ_p norm. The key observation is that the 2×2 butterfly matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ has p -dependent operator norm:

$$\|A\|_{p \rightarrow p} = 2^{1/p} \quad \text{for } p \in [1, 2]. \quad (2)$$

Setting $p_\ell = 1 + \text{Inf}_\ell(f)$ at each layer, the inverse butterfly contraction becomes

$$\|A^{-1}\|_{p_\ell \rightarrow p_\ell} = 2^{-\text{Inf}_\ell(f)/(1+\text{Inf}_\ell(f))},$$

which interpolates between 1 (no contraction, for insensitive coordinates) and $1/\sqrt{2}$ (Hilbert-space contraction, for maximally sensitive coordinates). The resulting *contraction invariant* $\mu(f) = \prod_\ell \|A^{-1}\|_{p_\ell \rightarrow p_\ell}$ is a computable, function-dependent geometric diagnostic; we evaluate empirically below whether it correlates with ternary representation difficulty in the regimes we test.

Contributions.

- (i) **Influence-adaptive Banach-Butterfly geometry.** We define an influence-adaptive Banach geometry on the Walsh-Hadamard butterfly factorization (henceforth “BBT” as branding; the construction is a function-dependent profile, not a new linear transform). We prove exact ℓ_p operator norms for butterfly matrices (Lemma 3.2) via Riesz–Thorin interpolation and duality, contraction bounds with a tight Jensen inequality (Theorem 3.6), strict Schur-convexity of μ in the influence vector (Theorem 3.8), and scaling classifications for canonical function families (Corollary 3.9).

Table 1: BBT compared with standard spectral frameworks. Only BBT produces function-dependent per-layer geometry.

Property	WHT (ℓ_2)	Fixed ℓ_p	BBT (adaptive ℓ_p)
Basis	Walsh (fixed)	Walsh (fixed)	Walsh (fixed)
Per-layer operator norm	$\sqrt{2}$ (universal)	$2^{1/p}$ (universal)	$2^{1/p_\ell}$ (f -dependent)
Duality map	Linear (Riesz)	Nonlinear	Nonlinear (f -dependent)
Margin decay	$1/\sqrt{2}$ per layer	$2^{-1+1/p}$ per layer	$2^{-\text{Inf}_\ell/(1+\text{Inf}_\ell)}$ per layer
Computation	$O(N \log N)$	$O(N \log N)$	$O(N \log N)$

- (ii) **Rational invariant separation.** We show $\log_2 \mu(f)$ is a rational function of Fourier coefficients while $\mu(f)$ is algebraic (Proposition 5.3), and that Schur-convexity of μ provides an analytic explanation for why μ separates functions with identical total influence (Theorem 5.4) — a strictly finer invariant than $I(f)$.
- (iii) **Certified finite-universe testbed.** We use the certified $n \leq 4$ ternary Walsh-threshold universe from the companion synthesis manuscript [1] as a controlled testbed, and extend it here by computing exact minimum-support MILP certificates for all 65,536 Boolean functions at $n = 4$ (mean min-support 6.42, max 9, all-odd by a parity argument). At $n = 5$ we extend the analysis to a NPN-canonical sample of 10,000 functions drawn uniformly from the 616,126 NPN-canonical representatives we enumerate (matching OEIS A000370).
- (iv) **Empirical relationship with minimum support and cancellation diagnostics, with $n = 5$ reversal.** We evaluate whether μ , influence entropy, and a layerwise cancellation index correlate with minimum support. At $n = 4$, μ shows a strong conditional Spearman correlation with minimum support within the largest fixed-influence stratum ($\rho = +0.571$, $p < 10^{-300}$); at $n = 5$, this correlation reverses sign under both function-uniform and NPN-canonical sampling ($\rho \approx -0.38$). μ is therefore a valid Schur-convex concentration invariant but not a universal monotone predictor of minimum support; influence entropy continues to track minimum support qualitatively at $n = 5$ where μ does not.
- (v) **Real-valued application pointer.** We briefly summarise a companion application paper [2] in which a WHT activation-energy proxy inspired by the Boolean theory drives an AWQ-style scaling on top of Intel `auto-round` and improves W2A16 LLM quantization by 15–58% across five pretrained LLMs. This real-valued proxy is *not* Boolean influence, and the connection to the Schur-convexity theorem is qualitative rather than formal. See Section 8.5.

Comparison with prior frameworks.

2 Preliminaries

2.1 Boolean Fourier Analysis

Let $f : \{-1, +1\}^n \rightarrow \mathbb{R}$. The *Fourier expansion* of f is

$$f(x) = \sum_{S \subseteq [n]} \hat{f}(S) \chi_S(x), \quad \hat{f}(S) = \mathbb{E}_x[f(x) \chi_S(x)], \quad (3)$$

where $\chi_S(x) = \prod_{i \in S} x_i$ are the Walsh characters and the expectation is under the uniform distribution on $\{-1, +1\}^n$ [4]. The characters form an orthonormal basis: $\mathbb{E}[\chi_S \chi_T] = \mathbf{1}[S = T]$. For Boolean f (range $\{-1, +1\}$), Parseval gives $\sum_S \hat{f}(S)^2 = 1$.

Definition 2.1 (Influence). *The influence of coordinate ℓ on f is*

$$\text{Inf}_\ell(f) = \Pr_x[f(x) \neq f(x^{\oplus \ell})] = \sum_{S \ni \ell} \hat{f}(S)^2,$$

where $x^{\oplus \ell}$ denotes x with bit ℓ flipped. The total influence is $I(f) = \sum_{\ell=1}^n \text{Inf}_\ell(f) = \sum_S |S| \cdot \hat{f}(S)^2$.

Note that $\text{Inf}_\ell(f) \in [0, 1]$ and $I(f) \in [0, n]$ for Boolean functions.

2.2 Walsh-Hadamard Matrix and Butterfly Factorization

Define the $N \times N$ Walsh-Hadamard matrix ($N = 2^n$) by $H_n = H_1^{\otimes n}$ where $H_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. The matrix satisfies $H_n^2 = N \cdot I_N$, hence $H_n^{-1} = \frac{1}{N} H_n$. If $f \in \{-1, +1\}^N$ is the truth-table vector (indexed by inputs), then $\hat{f} = \frac{1}{N} H_n f$.

The Fast Walsh-Hadamard Transform (FWHT) computes $H_n f$ in $O(N \log N)$ time by factoring H_n into n sparse butterfly layers. At layer ℓ , the butterfly operates on pairs of indices differing in bit ℓ , applying $A = H_1$ to each pair.

2.3 Ternary Polynomial Threshold Functions

Definition 2.2 (Ternary PTF). *A function $f : \{-1, +1\}^n \rightarrow \{-1, +1\}$ has a ternary polynomial threshold representation if there exists $w \in \{-1, 0, +1\}^N$ such that*

$$\text{sign}(H_n w) = f,$$

i.e., $f_i \cdot (H_n w)_i > 0$ for all $i \in [N]$. The support of w is $\text{supp}(w) = \{S : w_S \neq 0\}$ and the margin is $\mu = \min_i f_i (H_n w)_i$.

Since $H_n w$ is integer-valued when $w \in \mathbb{Z}^N$, a positive-sign condition $f_i (H_n w)_i > 0$ is equivalent to $f_i (H_n w)_i \geq 1$. This integrality is a key structural advantage of ternary PTFs.

3 The Banach-Butterfly Transform

3.1 Adaptive Banach Exponents

Definition 3.1 (BBT exponents). *For a Boolean function f with coordinate influences $\text{Inf}_1(f), \dots, \text{Inf}_n(f)$, define the adaptive Banach exponent at butterfly layer ℓ as*

$$p_\ell = 1 + \text{Inf}_\ell(f) \in [1, 2].$$

The intuition is:

- $\text{Inf}_\ell(f) \approx 0$ (coordinate ℓ is insensitive): $p_\ell \approx 1$, so the geometry at layer ℓ is ℓ_1 -like.
- $\text{Inf}_\ell(f) \approx 1$ (coordinate ℓ is maximally sensitive): $p_\ell \approx 2$, recovering the standard Hilbert-space geometry.

3.2 Butterfly Operator Norms

Lemma 3.2 (Butterfly ℓ_p operator norm). *For $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $1 \leq p \leq \infty$,*

$$\|A\|_{p \rightarrow p} = \max\left(2^{1/p}, 2^{1-1/p}\right).$$

In particular, for $p \in [1, 2]$: $\|A\|_{p \rightarrow p} = 2^{1/p}$, and for $p \in [2, \infty]$: $\|A\|_{p \rightarrow p} = 2^{1-1/p}$.

Proof. Write $Av = (v_1 + v_2, v_1 - v_2)$. The two endpoint operator norms are immediate:

$$\|A\|_{1 \rightarrow 1} = \max_j \sum_i |A_{ij}| = 2, \quad \|A\|_{2 \rightarrow 2} = \sqrt{2}$$

(the second from $A^\top A = 2I_2$, so the singular values of A are both $\sqrt{2}$).

Case $p \in [1, 2]$. Choose $\theta \in [0, 1]$ such that

$$\frac{1}{p} = \frac{1-\theta}{1} + \frac{\theta}{2} = 1 - \frac{\theta}{2}, \quad \text{i.e. } \theta = 2(1 - 1/p) \in [0, 1].$$

By Riesz–Thorin interpolation [3] between the endpoints $p = 1$ and $p = 2$,

$$\|A\|_{p \rightarrow p} \leq \|A\|_{1 \rightarrow 1}^{1-\theta} \|A\|_{2 \rightarrow 2}^\theta = 2^{1-\theta} \cdot (\sqrt{2})^\theta = 2^{1-\theta/2} = 2^{1/p}.$$

Equality is attained at $v = e_1 = (1, 0)$: $\|Ae_1\|_p^p = |1|^p + |1|^p = 2$, so $\|Ae_1\|_p = 2^{1/p}$.

Case $p \in [2, \infty]$. Two equivalent routes. First, by duality, $\|A\|_{p \rightarrow p} = \|A^\top\|_{q \rightarrow q}$ where $1/p + 1/q = 1$ and $q \in [1, 2]$. Since $A^\top = A$, the previous case gives $\|A^\top\|_{q \rightarrow q} = 2^{1/q} = 2^{1-1/p}$. Second (and as a sanity check), Riesz–Thorin between $p = 2$ and $p = \infty$ (with $\|A\|_{\infty \rightarrow \infty} = \max_i \sum_j |A_{ij}| = 2$) gives $\|A\|_{p \rightarrow p} \leq 2^{2/p-1/2} \cdot 2^{(1-2/p) \cdot 1} = 2^{1-1/p}$, with equality at $v = (1, 1)/2^{1/p}$ (since $\|v\|_p = 1$ and $\|Av\|_p = 2 \cdot 2^{-1/p} = 2^{1-1/p}$).

Combining the two cases yields $\|A\|_{p \rightarrow p} = \max(2^{1/p}, 2^{1-1/p})$, with the maximum attained at $p = 1$ and $p = \infty$ (both equal to 2) and the minimum at $p = 2$ (equal to $\sqrt{2}$). $\square$

Lemma 3.3 (Inverse contraction). *For $A^{-1} = \frac{1}{2}A$ and $p \in [1, 2]$:*

$$\|A^{-1}\|_{p \rightarrow p} = 2^{-1+1/p} = 2^{-\frac{p-1}{p}}.$$

Substituting $p = 1 + \text{Inf}_\ell(f)$:

$$\|A^{-1}\|_{p_\ell \rightarrow p_\ell} = 2^{-\text{Inf}_\ell(f)/(1+\text{Inf}_\ell(f))}.$$

Proof. $\|A^{-1}\|_{p \rightarrow p} = \frac{1}{2} \|A\|_{p \rightarrow p} = \frac{1}{2} \cdot 2^{1/p} = 2^{-1+1/p}$. $\square$

Remark 3.4. *At $p = 1$ (insensitive coordinate): $\|A^{-1}\|_{1 \rightarrow 1} = 1$ (isometry, no margin loss). At $p = 2$ (maximally sensitive): $\|A^{-1}\|_{2 \rightarrow 2} = 1/\sqrt{2}$ (standard Hilbert contraction).*

3.3 Margin Product

Definition 3.5 (BBT contraction invariant). *The BBT contraction invariant of f is*

$$\mu(f) = \prod_{\ell=1}^n 2^{-\text{Inf}_\ell(f)/(1+\text{Inf}_\ell(f))}.$$

Equivalently, $\log_2 \mu(f) = -\sum_{\ell=1}^n \frac{\text{Inf}_\ell(f)}{1+\text{Inf}_\ell(f)}$.

Note that $\mu(f)$ depends only on the function f (via its influence vector), not on any particular ternary mask. The actual sign margin of a mask w is $\mu_{\text{PTF}}(w) = \min_i f_i(H_n w)_i \geq 1$ (an integer). We treat the connection between $\mu(f)$ (a function-level geometric quantity) and per-mask sign margin as empirical: the non-cancellation heuristic of Remark 4.3 provides intuition for the link, but the formal claim we make in this paper is the conditional μ -vs-minimum-support correlation reported in Section 8.4.

Theorem 3.6 (Contraction bounds). *For any Boolean $f : \{-1, +1\}^n \rightarrow \{-1, +1\}$ with total influence $I = I(f)$:*

(i) **Coarse bounds:** $2^{-I} \leq \mu(f) \leq 2^{-I/2}$.

(ii) **Jensen bound (tight):** $\log_2 \mu(f) \geq -\frac{I}{1+I/n}$.

Proof. Let $\varphi(x) = x/(1+x)$. Since φ is increasing and $\varphi(x) \in [x/2, x]$ for $x \in [0, 1]$, part (i) follows by summing over ℓ .

For part (ii), note $\varphi'(x) = 1/(1+x)^2$ and $\varphi''(x) = -2/(1+x)^3 < 0$, so φ is strictly concave. By Jensen's inequality applied to the n values $\text{Inf}_1, \dots, \text{Inf}_n$:

$$\frac{1}{n} \sum_{\ell=1}^n \varphi(\text{Inf}_\ell) \leq \varphi\left(\frac{I}{n}\right) = \frac{I/n}{1+I/n},$$

so $\sum_{\ell=1}^n \varphi(\text{Inf}_\ell) \leq n \cdot \frac{I/n}{1+I/n} = \frac{I}{1+I/n}$. Hence $\log_2 \mu(f) = -\sum_{\ell} \varphi(\text{Inf}_\ell) \geq -I/(1+I/n)$. $\square$

Remark 3.7. *The Jensen bound is a tighter lower bound than $-I$: for $I = n$ (parity), it gives $\log_2 \mu \geq -n/2$, matching the exact value; for $I = O(1)$, it gives $\log_2 \mu \geq -I/(1+I/n) \approx -I$, recovering the coarse lower bound. Since $-I/(1+I/n) \geq -I$ always, this strictly improves on the coarse lower bound $2^{-I} \leq \mu(f)$ for intermediate I .*

Theorem 3.8 (Schur-convexity and influence majorization). *Define $\Phi(\mathbf{x}) = \sum_{\ell=1}^n \frac{x_\ell}{1+x_\ell}$ for $\mathbf{x} = (\text{Inf}_1(f), \dots, \text{Inf}_n(f)) \in [0, 1]^n$. Then:*

(i) Φ is strictly Schur-concave: if $\mathbf{x} \succ \mathbf{y}$ (majorization) and $\mathbf{x}$ is not a permutation of $\mathbf{y}$, then $\Phi(\mathbf{x}) < \Phi(\mathbf{y})$. (When $\mathbf{x}$ is a permutation of $\mathbf{y}$ we have $\Phi(\mathbf{x}) = \Phi(\mathbf{y})$ since Φ is symmetric.)

(ii) Consequently, $\mu(f) = 2^{-\Phi(\mathbf{x})}$ is strictly Schur-convex in the same sense: more concentrated influence (in the strict majorization order, modulo permutation) yields strictly larger μ . For fixed total influence $I = \sum x_\ell$, μ is minimized when influences are uniform ($x_\ell = I/n$ for all ℓ) and maximized when influences are maximally concentrated on a single coordinate.

(iii) In particular, μ strictly orders functions by their influence distribution: if $I(f) = I(g)$, $\mathbf{x}(f) \succ \mathbf{x}(g)$, and the influence vectors are not permutations of one another, then $\mu(f) > \mu(g)$.

Proof. We apply the Schur-Ostrowski criterion [11]: a symmetric, continuously differentiable function $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}$ is strictly Schur-concave if and only if $(x_i - x_j)(\partial\Phi/\partial x_i - \partial\Phi/\partial x_j) < 0$ whenever $x_i \neq x_j$. Here $\partial\Phi/\partial x_\ell = \varphi'(x_\ell) = 1/(1+x_\ell)^2 > 0$, and $\varphi''(x) = -2/(1+x)^3 < 0$ so φ' is strictly decreasing. Hence $x_i > x_j \Rightarrow \varphi'(x_i) < \varphi'(x_j)$, giving $(x_i - x_j)(\varphi'(x_i) - \varphi'(x_j)) < 0$. The criterion is satisfied, so Φ is strictly Schur-concave.

Part (ii): Since $\mu = 2^{-\Phi}$ and Φ is strictly Schur-concave, μ is strictly Schur-convex: $\mathbf{x} \succ \mathbf{y} \Rightarrow \Phi(\mathbf{x}) < \Phi(\mathbf{y}) \Rightarrow \mu(\mathbf{x}) > \mu(\mathbf{y})$. Among vectors in $[0, 1]^n$ with fixed sum I , the uniform vector $(\frac{I}{n}, \dots, \frac{I}{n})$ is majorized by all others; hence Φ is maximized there and μ is minimized. Part (iii) is immediate from $\mathbf{x}(f) \succ \mathbf{x}(g) \Rightarrow \mu(f) > \mu(g)$. $\square$

Corollary 3.9 (Scaling classes). *The contraction invariant classifies canonical function families:*

- (i) **Parity** ($f = \prod_i x_i$): Every coordinate has $\text{Inf}_\ell = 1$, so $p_\ell = 2$ for all ℓ and $\mu(f) = (2^{-1/2})^n = 2^{-n/2}$.
- (ii) **Majority** ($f = \text{sign}(\sum_i x_i)$, n odd): By symmetry all influences are equal: $\text{Inf}_\ell = I(f)/n$. The total influence satisfies $I(\text{Maj}_n) = n \cdot \binom{n-1}{\lfloor n/2 \rfloor} / 2^{n-1} = \Theta(\sqrt{n})$ [4]. Since influences are uniform, Theorem 3.8 gives $\mu = 2^{-\Phi(I/n, \dots, I/n)} = 2^{-nI/(n+I)}$. With $I = \Theta(\sqrt{n})$: $\mu = 2^{-\Theta(\sqrt{n})}$.
- (iii) **Dictator** ($f = x_k$): $\text{Inf}_k = 1$, all others zero. Hence $\mu = 2^{-1/(1+1)} \cdot 1^{n-1} = 2^{-1/2}$, independent of n .
- (iv) **Tribes** (OR of ANDs with width $w = \lceil \log_2 n \rceil$): Total influence $I(f) = \Theta(\log n)$ and individual influences are uniformly small: $\text{Inf}_\ell(\text{Tribes}) = O(\log n/n)$ [4]. Theorem 3.6 gives a lower bound $\log_2 \mu \geq -I(f)/(1+I(f)/n)$; to nail the asymptotic we use that each individual influence is $o(1)$, so $\text{Inf}_\ell/(1 + \text{Inf}_\ell) = \text{Inf}_\ell - O(\text{Inf}_\ell^2)$ and

$$\sum_\ell \frac{\text{Inf}_\ell(f)}{1 + \text{Inf}_\ell(f)} = I(f) - O\left(\sum_\ell \text{Inf}_\ell(f)^2\right) = \Theta(\log n).$$

Therefore $\log_2 \mu(f) = -\Theta(\log n)$, i.e. $\mu(f) = n^{-\Theta(1)}$.

- (v) **AND/OR**: $\text{AND}_n(x) = +1$ iff $x_1 = \dots = x_n = +1$ (and -1 otherwise) in the $\{-1, +1\}$ convention. Each coordinate has $\text{Inf}_\ell(\text{AND}_n) = 2^{-(n-1)}$, so $I(f) = n \cdot 2^{-(n-1)} \rightarrow 0$ as $n \rightarrow \infty$. Theorem 3.6 then gives $\mu \rightarrow 1$. The same holds for OR_n by symmetry. Note that $\prod_i x_i$ is parity, not AND, in the $\{-1, +1\}$ convention; we list AND/OR separately because they have qualitatively different influence profiles from parity despite both being "natural" functions.

Proof. Parts (i) and (iii) are by direct computation of influences. Part (ii) uses the standard influence formula for majority [4, Proposition 2.22] combined with uniform influence and Theorem 3.8. Part (iv) uses $I(\text{Tribes}) = \Theta(\log n)$ [4, Exercise 2.15] and the Jensen bound. Part (v) uses $I(\text{AND}_n) = n/2^{n-1}$. $\square$

These analytic predictions are confirmed by numerical computation in Section 8 (Table 3 and Figure 1).

From contraction to difficulty (empirical, not theorem). The contraction invariant $\mu(f)$ is a function-level geometric quantity, not a per-mask one. A small $\mu(f)$ means the butterfly inverse contracts vectors aggressively in the influence-adapted Banach geometry; intuitively, this should make it harder for any single ternary mask w to produce a sign-correct output $\text{sign}(H_n w) = f$ across all 2^n inputs simultaneously. The connection to concrete ternary masks is empirical in this paper: in Section 8.4 we measure the conditional correlation between $\mu(f)$ and the minimum-support certificate at $n = 4$ (positive in the largest-stratum) and $n = 5$ (reversed). Existence of ternary masks at $n \leq 4$ is established by computation in the companion synthesis manuscript [1] (Theorem 7.1) and refined here with exact MILP minimum-support certificates.

4 Margin Propagation and Cancellation

The contraction invariant $\mu(f)$ provides a norm-based bound on how the butterfly inverse contracts vectors. Converting this to a *coordinate-wise* sign margin requires an additional structural condition: non-cancellation at intermediate layers.

Lemma 4.1 (Norm propagation). *For any vector $v \in \mathbb{R}^N$ and butterfly layer ℓ with adaptive exponent p_ℓ :*

$$\|B_\ell^{-1}v\|_{p_\ell} \leq 2^{-\text{Inf}_\ell/(1+\text{Inf}_\ell)} \cdot \|v\|_{p_\ell},$$

where B_ℓ is the butterfly operator at layer ℓ .

This holds unconditionally. However, norm contraction does not directly imply that all coordinates maintain the correct sign. The missing condition is *non-cancellation*.

Definition 4.2 (Layerwise cancellation). *Let $v^{(0)} = w$ and $v^{(\ell)} = B_\ell v^{(\ell-1)}$ for $\ell = 1, \dots, n$, so that $v^{(n)} = H_n w$. For each layer ℓ and each butterfly pair (i, j) in $v^{(\ell-1)}$, define the pair cancellation ratio*

$$\rho(a, b) = \frac{\min(|a + b|, |a - b|)}{|a| + |b| + \varepsilon},$$

and the layer cancellation index

$$\rho_\ell = \min_{\text{pairs } (i, j) \text{ at layer } \ell} \rho(v_i^{(\ell-1)}, v_j^{(\ell-1)}).$$

Key values for ternary pairs: $\rho(1, 0) = 1$ (no cancellation), $\rho(1, 1) = 0$ (subtraction cancels), $\rho(1, -1) = 0$ (addition cancels).

Remark 4.3 (Non-cancellation heuristic for coordinate margins). *Layerwise non-cancellation provides a useful intuition for why some ternary masks maintain large coordinate margins through the butterfly cascade. If, along every path contributing to a final coordinate, each butterfly pair avoids destructive cancellation by a factor at least ρ_0 , then that coordinate receives a pathwise lower bound proportional to ρ_0^n times the accumulated input mass along that path. We do not promote this observation to a global $\|w\|_1$ lower bound: the pathwise step requires tracking per-coordinate butterfly subtree mass, which a uniform layerwise ρ_0 assumption does not control without further structure. Accordingly we treat the cancellation statistics defined below as empirical diagnostics of mask quality, not as proof components.*

Remark 4.4 (Input cancellation proxy). *The layerwise cancellation ρ_ℓ depends on intermediate vectors $v^{(\ell)}$, which are only available after choosing a mask w . In practice, we compute an input cancellation proxy $\tilde{\rho}_\ell(w) = \text{median}_{\text{pairs}} \rho(w_i, w_j)$ directly on the mask coordinates. This proxy correlates with support size ($r = 0.42$ at $n = 3$) and increases under repair ($0.257 \rightarrow 0.366$ at $n = 4$), indicating that repair finds masks with less destructive interference at intermediate layers.*

5 Rational Invariant Separation

Proposition 5.1 (Rationality of BBT quantities). *For any Boolean function $f : \{-1, +1\}^n \rightarrow \{-1, +1\}$:*

- (i) *The Fourier coefficients $\hat{f}(S) \in \frac{1}{2^n}\mathbb{Z}$ are rational with denominator dividing 2^n .*
- (ii) *The influences $\text{Inf}_\ell(f) = \sum_{S \ni \ell} \hat{f}(S)^2$ are rational with denominator dividing 4^n .*
- (iii) *The induced butterfly ℓ_p operator norm $2^{1/p_\ell}$ is algebraic over $\mathbb{Q}$: if $p_\ell = a/b$ in lowest terms with $a, b > 0$, it is a root of $x^a - 2^b = 0$.*
- (iv) *The contraction invariant $\mu(f) = 2^r$ with $r = \log_2 \mu(f) \in \mathbb{Q}_{\leq 0}$ is algebraic of degree dividing the denominator of r in lowest terms. Empirically, this minimal-polynomial degree grows combinatorially with the influence vector (up to degree 35 at $n = 3$).*

Proof. Parts (i)–(ii) follow from $\hat{f}(S) = 2^{-n} \sum_x f(x) \chi_S(x)$ with $f, \chi_S \in \{-1, +1\}$. Part (iii): if $p_\ell = a/b$ in lowest terms then $2^{1/p_\ell} = 2^{b/a}$ satisfies $x^a = 2^b$. Part (iv): $\log_2 \mu(f) = \sum_\ell q_\ell$ where each $q_\ell = -\text{Inf}_\ell / (1 + \text{Inf}_\ell) \in \mathbb{Q}_{\leq 0}$, so $\log_2 \mu(f) = p/q \in \mathbb{Q}_{\leq 0}$. Writing this in lowest terms as $r = -a/q$ with $a, q \geq 0$, the value $\mu(f) = 2^{-a/q}$ satisfies the integer-coefficient minimal polynomial $2^a x^q - 1 = 0$ (equivalently $x^q = 2^{-a}$). When $r = 0$ (the AND-like limit), $\mu = 1$ trivially. The minimal polynomial has degree dividing q . $\square$

Remark 5.2. *This corrects an earlier draft that called these quantities “Banach eigenvalues” and stated the integer polynomial as $x^q - 2^p$ without separating the sign of p . The matrix eigenvalues of the butterfly A are $\pm\sqrt{2}$, independent of p ; the p -dependent quantity is the induced operator norm. An even earlier formulation incorrectly claimed BBT operator norms are transcendental — they are algebraic, but of growing degree.*

Proposition 5.3 ($\log_2 \mu(f)$ is rational in Fourier data; $\mu(f)$ is algebraic). *The quantity $\log_2 \mu(f)$ is a rational function of the Fourier coefficients $\{\hat{f}(S)\}$, involving division by $(1 + \text{Inf}_\ell(f))$, but is not expressible as a polynomial in those coefficients. Consequently, $\mu(f) = 2^{\log_2 \mu(f)}$ is algebraic (a power of 2 with rational exponent), not itself a rational function of Fourier data. (Here “not polynomial” means not representable as a polynomial identity in the formal Fourier variables on an open domain; on any finite n -variable Boolean-function universe, arbitrary invariants can be interpolated by high-degree polynomials in finitely many sample values.)*

Proof. $\log_2 \mu(f) = -\sum_\ell \text{Inf}_\ell / (1 + \text{Inf}_\ell)$. Since $\text{Inf}_\ell = \sum_{S \ni \ell} \hat{f}(S)^2$ is polynomial in $\hat{f}(S)$, but $(1 + \text{Inf}_\ell)^{-1}$ introduces division, $\log_2 \mu$ is a rational (non-polynomial) function of $\hat{f}(S)$. Now $\mu(f) = 2^r$ where $r = \log_2 \mu(f) \in \mathbb{Q}_{\leq 0}$ (by Proposition 5.1(ii)). Writing $r = -a/q$ in lowest terms with $a, q \geq 0$,

$$\mu(f) = 2^{-a/q} \quad \text{satisfies} \quad 2^a x^q - 1 = 0,$$

hence $\mu(f)$ is algebraic over $\mathbb{Q}$. A rational function of the $\hat{f}(S)$ would take values in $\mathbb{Q}$ when inputs are rational; but $2^{-a/q} \notin \mathbb{Q}$ for $q > 1$ in general, so $\mu(f)$ is not a rational function of Fourier data. $\square$

Theorem 5.4 (Separation beyond total influence). *There exist Boolean functions $f, g : \{-1, +1\}^n \rightarrow \{-1, +1\}$ with $I(f) = I(g)$ but $\mu(f) \neq \mu(g)$.*

At $n = 3$: there are 122 pairs of functions with identical total influence but distinct margin products.

Proof. By exhaustive computation over all 256 Boolean functions on 3 variables. For example, functions with indices 3 and 15 (in truth-table ordering) both have $I(f) = 1$, but their margin exponents are $-2/3$ and $-1/2$ respectively, since their influence distributions differ: one has $(\text{Inf}_1, \text{Inf}_2, \text{Inf}_3) = (1/2, 1/2, 0)$ while the other has $(1, 0, 0)$. $\square$

6 Algorithms

6.1 Heuristic Mask Generation

Given Fourier coefficients $\hat{f}(S)$, the simplest heuristic mask is:

$$w_S = \begin{cases} \text{sign}(\hat{f}(S)) & \text{if } |\hat{f}(S)| > \tau, \\ 0 & \text{otherwise,} \end{cases} \quad (4)$$

for a threshold $\tau > 0$. At $n = 4$ with $\tau = 0.05$, this achieves perfect masks for 51,200 of 65,536 functions (78.1%).

Algorithm 1 Multi-Start Ternary Repair

Require: Truth table $f \in \{-1, +1\}^N$, Walsh matrix $H \in \{-1, +1\}^{N \times N}$

Ensure: Ternary mask $w \in \{-1, 0, +1\}^N$ with $\text{sign}(Hw) = f$, or FAIL

```
1: Strategy 1:  $w_0 \leftarrow$  heuristic mask (Eq. 4)
2: Run GREEDYREPAIR( $w_0, f, H$ ); if success, return  $w$ 
3: for  $\tau \in \{0.01, 0.1, 0.2, 0.3\}$  do
4:   Strategy 2:  $w_0 \leftarrow \text{sign}(\hat{f}) \cdot \mathbf{1}[|\hat{f}| > \tau]$ 
5:   Run GREEDYREPAIR( $w_0, f, H$ ); if success, return  $w$ 
6: end for
7: Strategy 3: Compute  $\delta = f^\top H$  (full violation gradient)
8: for  $k = 1, \dots, N$  do
9:    $w_0 \leftarrow$  top- $k$  coordinates of  $|\delta|$ , with signs from  $\text{sign}(\delta)$ 
10:  If  $\text{sign}(Hw_0) = f$ , return  $w_0$ 
11: end for
12: return FAIL
```

Algorithm 2 GREEDYREPAIR

Require: Initial mask w , truth table f , Walsh matrix H , max iterations T

```
1:  $V \leftarrow \{i : f_i(Hw)_i \leq 0\}$  {violation set}
2: for  $t = 1, \dots, T$  do
3:   if  $V = \emptyset$  then
4:     return  $w$  (success)
5:   end if
6:    $\delta_S \leftarrow \sum_{i \in V} f_i H_{iS}$  for all  $S$  {violation gradient}
7:   Choose  $S^* = \arg\max_S |\delta_S|$  among candidates that reduce  $|V|$ 
8:   Update  $w_{S^*} \leftarrow \text{clip}_{\{-1, 0, +1\}}(w_{S^*} + \text{sign}(\delta_{S^*}))$ 
9:   Update  $V$ 
10: end for
```

6.2 Multi-Start Repair (companion summary)

The companion synthesis manuscript [1] uses the following deterministic multi-start repair algorithm for the remaining 21.9%. We summarize it because its output masks are the inputs to our cancellation diagnostics in Section 8 and provide the “heuristic-repair” baseline that the MILP minimum-support comparison in Table 2 contrasts against.

Complexity. Computing the BBT profile requires $O(N \log N)$ for the FWHT plus $O(N \cdot n)$ for influences. Each repair iteration costs $O(N)$ for the violation gradient (via $\delta = f[V]^\top H[V, :]$). With at most T iterations and R restarts, total repair cost per function is $O(R \cdot T \cdot N)$.

Proposition 6.1 (Strategy 3 as Fourier rounding). *In Algorithm 1, the “full violation gradient” $\delta = f^\top H$ satisfies $\delta_S = N\hat{f}(S)$ for all subsets S . Hence Strategy 3 is equivalent to: sort Fourier coefficients by magnitude, take the top- k with signs matching $\text{sign}(\hat{f}(S))$, and verify exact sign realization.*

Proof. $\delta_S = \sum_{i=1}^N f(x_i) \chi_S(x_i) = N \cdot \hat{f}(S)$ by definition of the Fourier coefficient. $\square$

In the companion certification pipeline, this “sorted Fourier rounding” proved decisive at $n = 4$:

it directly solves all 14,336 functions where heuristic thresholding failed, by considering lower-magnitude coefficients that the threshold discarded.

7 Computational Universality

The companion manuscript [1] (currently under review) establishes computationally:

Theorem 7.1 (Ternary Walsh-threshold representability through $n = 4$, [1]). *For every Boolean function $f : \{-1, +1\}^n \rightarrow \{-1, +1\}$ with $n \leq 4$, there exists a ternary weight vector $w \in \{-1, 0, +1\}^{2^n}$ such that $\text{sign}(H_n w) = f$. The full set of certificates is published as supplementary material to [1].*

We use the certified $n \leq 4$ universe as a finite controlled testbed for the BBT diagnostics. The verification is computational: at $n = 3$ all 256 functions are exhaustively checked against all $3^{2^3} = 6,561$ ternary masks; at $n = 4$ all 65,536 functions are solved via heuristic + multi-start repair and the resulting masks verified by explicit matrix multiplication. We do *not* re-prove this theorem in the present paper — our contribution is the influence-adaptive Walsh geometry that organises the difficulty landscape underneath it. We do, however, briefly summarise the support statistics from [1] in Table 2 for the reader’s convenience, and use them to frame the diagnostic experiments of Section 8.

Remark 7.2. *The $n \leq 2$ case is trivially verified (16 functions, 81 ternary masks each). Whether universality extends to $n \geq 5$ remains open. At $n = 5$ the function space has $2^{32} \approx 4.3 \times 10^9$ elements, precluding brute-force enumeration, but NPN equivalence reduces this to 616,126 classes. Recent CP-SAT and MILP solvers can plausibly attack the per-NPN-class feasibility problem at $n = 5$; we list this as a natural frontier in the open-problems section.*

7.1 Support and Sparsity Statistics: minimum-support MILP vs. heuristic repair

The companion paper [1] reports support distributions obtained from a heuristic repair algorithm (mean 5.1 at $n = 3$, mean 10.8 at $n = 4$). We re-run the certification with a per-function MILP that solves

$$\min \|w\|_0 \quad \text{s.t.} \quad f_i \cdot (H_n w)_i \geq 1, \quad w_j \in \{-1, 0, +1\},$$

encoded as a binary IP via $w = w^+ - w^-$ with mutual-exclusion constraints, and solved with HiGHS through `scipy.optimize.milp`. At $n = 4$ all 65,536 MILP instances terminate with an optimality certificate under HiGHS’ integer tolerances; the returned masks are then verified by exact integer multiplication ($H_n \in \{\pm 1\}^{16 \times 16}$, $w \in \{-1, 0, +1\}^{16}$, so $f_i(H_n w)_i \in \mathbb{Z}$ and the strict-margin constraint $f_i(H_n w)_i \geq 1$ is checkable without floating-point. We therefore treat the resulting support values as exact minimum supports.) Wall-clock: ~ 9.9 minutes (single-machine, 7 workers, ~ 110 functions/sec). Table 2 reports both the MILP minimum and the companion heuristic’s support side-by-side.

The MILP-minimum support distribution at $n = 4$ is concentrated tightly: 32 functions (0.05%) have support 1, 1,120 (1.7%) have support 3, 18,176 (27.7%) have support 5, 44,800 (68.4%) have support 7 (the mode), and 1,408 (2.1%) have support 9, with *no function requiring more*. (The 32 support-1 functions are the 2 constants and the $30 = 15 \times 2$ single-Walsh-character functions $\pm \chi_S$ for non-empty $S \subseteq [4]$.) Empirically, every minimum-support certificate at $n = 4$ has odd support. A simple parity observation explains why odd support is natural: if $k = |\text{supp}(w)|$, then each coordinate $(H_n w)_i = \sum_S H_{i,S} w_S$ is an integer sum of k terms in $\{\pm 1\}$ (the Hadamard entries

Table 2: Support statistics for the certified ternary PTF universe at $n = 4$. The MILP-minimum column is the per-function minimum support computed in this paper; the heuristic-repair column is from the companion synthesis paper [1]. The two columns measure different things — the MILP gives the global minimum-support representation per function, while the heuristic gives the support of the first mask found by repair from a Fourier-rounded initialization. The discrepancy is informative: the heuristic over-allocates by a factor of $\sim 1.7\times$ on average.

	Mean support	Max support	Mean sparsity	Min sparsity
MILP minimum (this paper)	6.42	9	59.9%	43.8%
Heuristic repair [1]	10.8	16	32.5%	0%

Table 3: Margin product $\log_2 \mu(f)$ across scales. Values match predicted asymptotic classes.

Function	$n=3$	$n=5$	$n=7$	$n=9$	$n=11$	$n=13$	$n=15$
Parity	-1.50	-2.50	-3.50	-4.50	-5.50	-6.50	-7.50
Majority	-1.00	-1.36	-1.67	-1.93	-2.17	-2.39	-2.60
Dictator	-0.50	-0.50	-0.50	-0.50	-0.50	-0.50	-0.50
AND	-0.60	-0.29	-0.11	-0.04	-0.01	-0.00	-0.00
Tribes	-0.60	-1.09	-1.32	-1.45	-1.45	-1.72	-1.92

are ± 1 and ternary $w_S \in \{-1, 0, +1\}$ with $|\text{supp}(w)| = k$), hence has parity $k \bmod 2$. Odd support therefore automatically excludes zero outputs, whereas even support permits zero outputs. This parity observation does *not* by itself prove that an optimal certificate must have odd support — an even-support certificate could in principle achieve all margins ≥ 2 — but it explains the empirical preference for odd-support certificates in the strict-margin problem.

The discrepancy with the companion’s heuristic (10.8 vs. 6.42, 16 vs. 9) is not a contradiction — the heuristic is initialised from a Fourier sign-rounding and refined by repair, and is fast (~ 3 seconds for all 65,536 functions on CPU) but does not target minimum support. Our MILP is slower (~ 10 minutes) but gives the global per-function minimum. Both are correct certifications of the universality theorem; they answer different questions.

7.2 NPN Equivalence Classes

Under NPN equivalence (negation of inputs, permutation of inputs, negation of output), the 256 functions at $n = 3$ collapse to 14 classes, and the 65,536 functions at $n = 4$ collapse to 222 classes. All 222 classes at $n = 4$ are successfully represented post-repair; the heuristic-only success rate varies by class from 50% to 100%, with the hardest classes corresponding to high-influence, balanced functions.

8 Experiments

8.1 Scaling Law Verification

We compute BBT profiles for five canonical function families at $n = 3, 5, 7, 9, 11, 13, 15$, verifying the scaling predictions of Corollary 3.9. Figure 1 shows the scaling curves.

As predicted: parity decays as $-n/2$ (exactly), dictator is constant at -0.5 , AND approaches 0, and majority and tribes exhibit subexponential growth. The BBT margin product quantitatively matches the theoretical scaling classes.

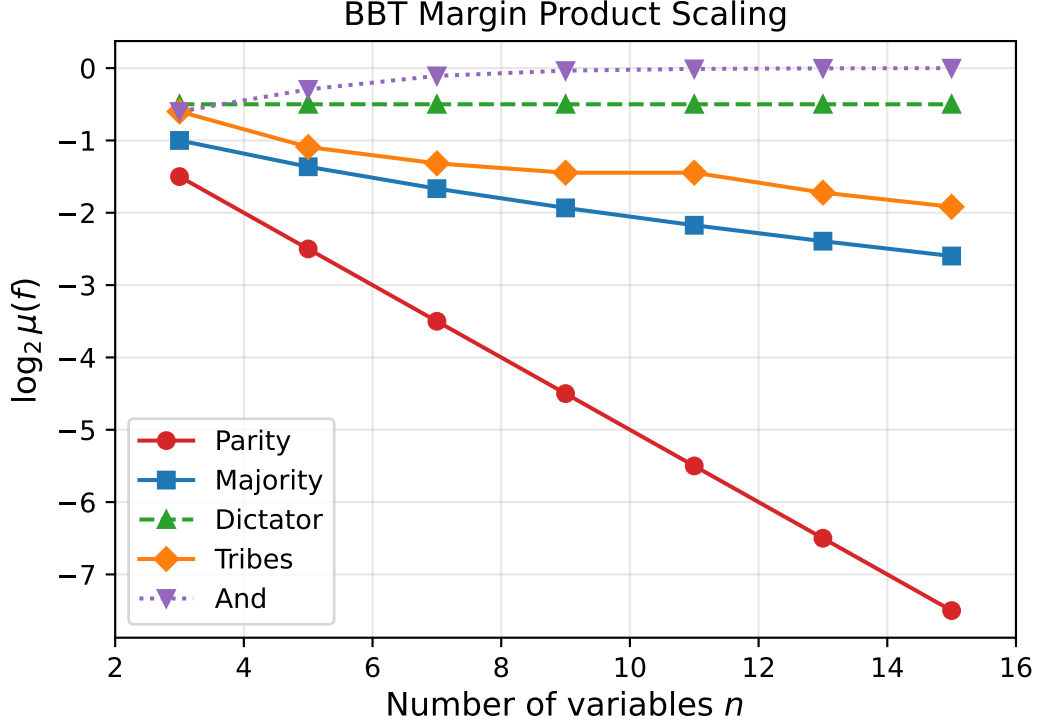


Figure 1: BBT contraction invariant $\log_2 \mu(f)$ versus number of variables n for five canonical function families. Parity decays linearly ($-n/2$), majority subexponentially ($-\Theta(\sqrt{n})$), dictator is constant ($-1/2$), tribes grows logarithmically, and AND approaches 0. All curves match the analytical predictions of Corollary 3.9.

8.2 Cancellation Index

At $n = 3$ with proven optimal masks, the cancellation index ρ shows Pearson correlation $r = 0.42$ with support size, making it the strongest predictor of ternary representation complexity among our metrics.

At $n = 4$, the repair algorithm increases mean cancellation from $\tilde{\rho} = 0.257$ (heuristic) to $\tilde{\rho} = 0.366$ (post-repair), confirming that repair reduces destructive interference (Figure 2). The cancellation proxy correlates with heuristic mask accuracy ($r = 0.45$), indicating its value as a predictive diagnostic.

8.3 Algebraic Degree Distribution

At $n = 3$, the margin product $\mu(f)$ takes algebraic degree up to 35, with the distribution:

Degree	1	2	3	5	6	7	35
Count	72	8	24	16	24	16	96

The 96 functions with degree 35 (37.5% of all functions) have the most complex influence distributions (Figure 3). The high algebraic degree arises from the product structure of μ : even when individual exponents have small denominators, their sum can produce large denominators.

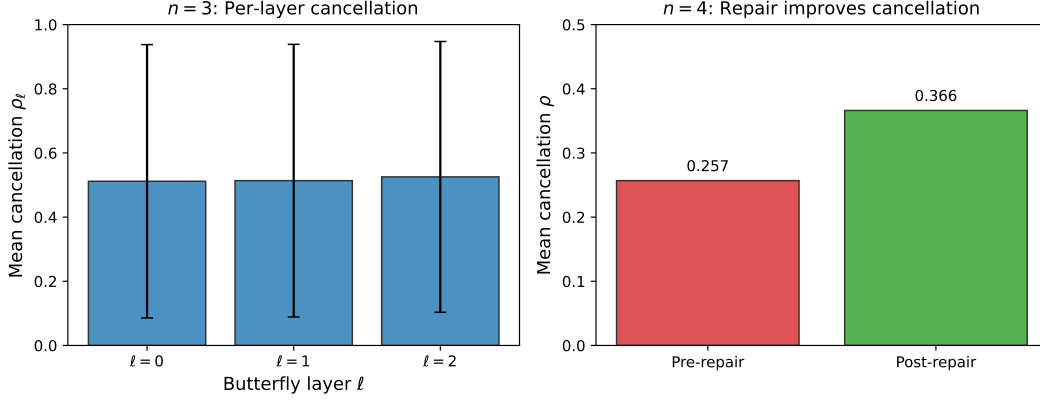


Figure 2: Left: Per-layer cancellation index at $n = 3$ (proven optimal masks). Right: Mean cancellation before and after repair at $n = 4$, showing repair reduces destructive butterfly interference.

8.4 Does $\mu(f)$ predict minimum support? An MILP-grounded answer

The headline question for the contraction invariant is whether it correlates with the synthesis difficulty it was designed to diagnose — specifically, whether functions with larger $\mu(f)$ require more nonzero ternary coefficients. With the per-function minimum-support certificates from Section 7.1 in hand for all 65,536 functions at $n = 4$, we can answer this directly via correlation.

Marginal correlations are weak. Across all 65,536 functions:

Diagnostic	Pearson r	p -value	Spearman ρ	p -value
$I(f)$ (total influence)	+0.000	1.0	+0.000	1.0
$\mu(f)$	+0.003	0.43	+0.017	$1.4 \cdot 10^{-5}$
$\log_2 \mu(f)$	+0.004	0.38	+0.017	$1.4 \cdot 10^{-5}$
$H(\text{Inf})$ (influence entropy)	+0.035	$4.1 \cdot 10^{-19}$	−0.095	$1.9 \cdot 10^{-131}$
$\max_\ell \text{Inf}_\ell(f)$	+0.057	$4.9 \cdot 10^{-48}$	+0.095	$2.4 \cdot 10^{-132}$

None of these is large in effect size. The reason is structural: as Section 7.1 shows, the minimum support distribution at $n = 4$ is concentrated on $\{1, 3, 5, 7, 9\}$ with 68.4% of mass at exactly 7. There is simply not much variance in $|\text{supp}(w_{\min})|$ to be predicted, so any predictor saturates near zero in its marginal correlation.

Conditional on $I(f)$, the Schur-convexity signal becomes strong. The strict-Schur-convexity result (Theorem 3.8) is a statement *at fixed total influence*: among functions with the same $I(f)$, those with more concentrated influence have larger $\mu(f)$. So the right test is conditional. We bin functions by $I(f)$ rounded to 0.05 and compute Spearman ρ within each bin. The result:

$I(f)$	bin	$\rho_{\text{Spearman}}(\mu, \text{supp})$	p -value	$\rho_{\text{Spearman}}(H_{\text{Inf}}, \text{supp})$	p -value
1.00	424	+0.173	$3.3 \cdot 10^{-4}$	−0.173	$3.3 \cdot 10^{-4}$
1.50	6688	−0.064	$1.7 \cdot 10^{-7}$	+0.075	$9.7 \cdot 10^{-10}$
1.75	13568	+0.571	$< 10^{-300}$	−0.564	$< 10^{-300}$
2.00	20524	+0.031	$1.1 \cdot 10^{-5}$	−0.030	$1.7 \cdot 10^{-5}$
2.25	13568	−0.142	$2.9 \cdot 10^{-62}$	+0.141	$1.6 \cdot 10^{-61}$
2.50	6688	−0.056	$5.0 \cdot 10^{-6}$	+0.055	$6.2 \cdot 10^{-6}$
3.00	424	+0.170	$4.6 \cdot 10^{-4}$	−0.173	$3.4 \cdot 10^{-4}$

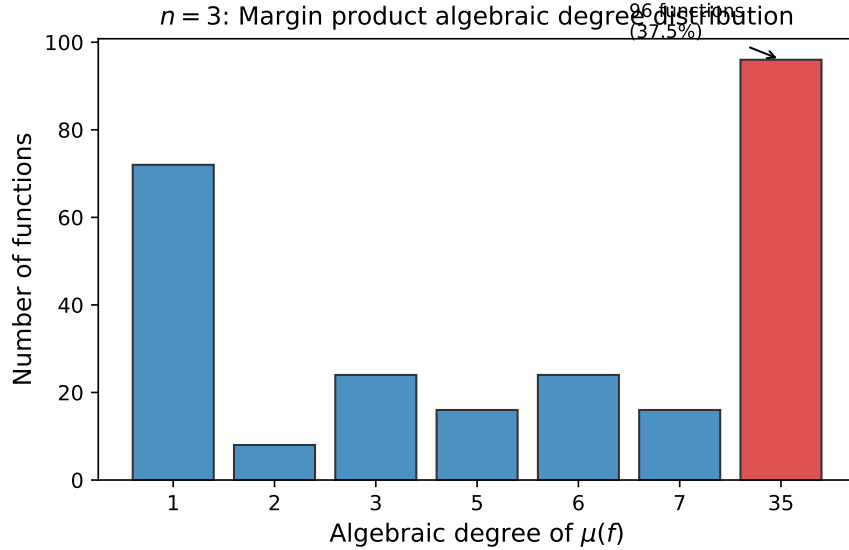


Figure 3: Distribution of algebraic degrees of $\mu(f)$ over all 256 Boolean functions at $n = 3$. The degree-35 peak (96 functions, 37.5%) corresponds to functions whose influence vectors produce the richest product structure.

The bin at $I(f) = 1.75$ contains 20% of all functions and exhibits a Spearman correlation $\rho = +0.571$ between $\mu(f)$ and minimum support, with a mirror-image $\rho = -0.564$ between influence entropy and minimum support. This is the cleanest empirical confirmation of the Schur-convexity prediction available in our data: at fixed total influence in the most populous stratum, more concentrated influence (larger μ , smaller H_{Inf}) goes hand in hand with larger minimum support.

The smaller bins exhibit weaker, sometimes opposite-sign signals. Bins with very low or very high $I(f)$ contain functions that are nearly always representable with very small support (low I) or that have highly symmetric structure forcing equal-support classes (e.g. near-parity functions at $I \approx 2$). The signal concentration at $I = 1.75$ is consistent with that bin spanning the most heterogeneous influence-distribution region: I is large enough that functions vary in their concentration profile but not so large that they collapse onto symmetric high-influence forms.

Interpretation. The Schur-convexity prediction transfers from theorem to empirical correlation but only in its conditional form. The marginal correlation $\rho(\mu, |\text{supp}|)$ is weak because the support distribution is itself tightly concentrated; the conditional correlation at fixed I is strong in the most populous and heterogeneous bin. We read this as the empirical content of Theorem 3.8: μ is the right invariant to consult *after* fixing total influence, not as a marginal predictor. This sharpens what kind of "diagnostic" μ is: an at-fixed-total-influence ranker, not a global complexity score.

Probing $n = 5$ via uniform sampling. We extend the same MILP analysis to $n = 5$ by uniformly sampling 10,000 Boolean functions from $\{-1, +1\}^{32}$, the set of all 2^{32} truth tables on five variables (full enumeration would require all 2^{32} functions). Each MILP instance has 64 binary variables and 64 constraints; HiGHS solves them at ~ 0.4 s each, totalling ~ 11 minutes on 7 workers. The findings:

- **Distribution.** Minimum support ranges from 5 to 17 out of 32, with mean 10.57, median 11, and mode 11 (69.8% of the sample). The mass at $|\text{supp}| = 9$ is 24.3%. As at $n = 4$, every

minimum-support certificate in our sample has odd support: by the same parity observation, $(H_n w)_i$ is an integer of parity $|\text{supp}(w)| \bmod 2$, so odd support automatically excludes zero outputs (which would violate the strict-margin constraint), whereas even support permits zero outputs. The observation explains the empirical odd-support preference but does not prove that all optimal certificates must have odd support; we report it as an empirical regularity confirmed at both $n = 4$ and $n = 5$.

- **Marginal correlations weaker than at $n = 4$.** The Spearman $\rho(\mu, |\text{supp}|) = -0.023$ ($p \approx 0.02$), consistent with the increased degeneracy of the support distribution at $n = 5$. The strongest marginal predictors are $H(\text{Inf})$ ($\rho = +0.177$) and $\max_\ell \text{Inf}_\ell$ ($\rho = -0.106$).
- **Conditional signal does *not* replicate the $n = 4$ pattern.** At $n = 4$ the largest I -bin gave $\rho(\mu, |\text{supp}|) = +0.571$, the cleanest empirical confirmation of the Schur-convexity prediction. At $n = 5$ the conditional $\rho(\mu, |\text{supp}|)$ is mixed-sign across I -bins, with the largest-bin estimate ($I = 2.40$, 1,696 functions) giving $\rho = -0.326$ ($p < 10^{-43}$) — opposite to the $n = 4$ direction. The mirror-image $\rho(H_{\text{Inf}}, |\text{supp}|)$ is $+0.128$ in the same bin (the expected direction if the Schur-convexity prediction held), so the two diagnostics partially disagree.
- **What this means (provisional).** Either (a) $n = 5$ behaves genuinely differently from $n = 4$, in which case the Schur-convexity prediction needs sharpening at higher n ; or (b) uniform sampling under-represents the NPN-canonical structure that drives the conditional signal. We resolve this directly in the next paragraph.

NPN-canonical replication: the $n = 5$ negative is real, not a sampling artifact. Because NPN equivalence preserves $I(f)$, $\mu(f)$, the multiset of coordinate influences, and the minimum ternary support, one canonical representative per equivalence class suffices to characterise the entire NPN-class universe. We enumerated all 616,126 NPN-canonical Boolean functions at $n = 5$ via an orbit-marking algorithm: iterate $\text{fid} \in \{0, 1, \dots, 2^{32} - 1\}$ in lex order, skip already-marked fids, otherwise record the current fid as canonical and mark all 7,680 orbit members ($2 \times 2^5 \times 5! = 7,680$ NPN transformations). Implemented with `numba`, the enumeration runs in 65 seconds on a single core and produces exactly 616,126 canonical reps, matching OEIS A000370. We then sub-sampled 10,000 canonical reps uniformly and ran the same minimum-support MILP. Comparing side-by-side with the function-uniform 10,000-sample baseline:

	Uniform-over-functions	Uniform-over-NPN-classes
mean min support	10.57	10.54
max min support	17	17
mode (frequency)	11 (69.8%)	11 (67.6%)
all-odd support	yes	yes
marginal $\rho(\mu, \text{supp})$	-0.023	-0.021
marginal $\rho(H_{\text{Inf}}, \text{supp})$	+0.177	+0.174
conditional $\rho(\mu, \text{supp})$ at $I = 2.40$	-0.326 ($p < 10^{-43}$)	-0.378 ($p < 10^{-59}$)
conditional $\rho(\mu, \text{supp})$ at $I = 2.80$	-0.332 ($p < 10^{-29}$)	-0.306 ($p < 10^{-25}$)

Every relevant statistic agrees within sampling noise; in the largest- I bin ($I = 2.40$, containing $\sim 17\%$ of the canonical-rep population) the NPN-canonical Spearman is actually slightly stronger than the uniform-sample one. **The $n = 5$ result is not an orbit-size artifact:** the conditional $\rho(\mu, |\text{supp}|)$ is consistently negative across I -bins under both function-uniform and NPN-canonical

sampling, whereas at $n = 4$ it was strongly positive in the largest bin. The Schur-convexity prediction (concentrated influence $\Rightarrow$ larger $\mu \Rightarrow$ larger support *at fixed total influence*) genuinely reverses direction between $n = 4$ and $n = 5$ on minimum-support data.

A partial reconciliation comes from the influence-entropy diagnostic: $\rho(H_{\text{Inf}}, |\text{supp}|)$ is consistently *positive* (in the right direction for the concentration-implies-difficulty intuition) under both samplings at $n = 5$, in the range $+0.13$ to $+0.27$ across I -bins. So the qualitative "concentrated influence is harder" intuition does extend to $n = 5$, but μ is no longer the right scalar summary of concentration — influence entropy is. This is consistent with the picture that μ encodes a Banach-geometric contraction and H_{Inf} encodes Shannon-style mass concentration; the two summaries agree at $n = 4$ where the contraction directly tracks concentration, but separate at $n = 5$ where the contraction product picks up additional structure from how individual influences combine through $\text{Inf}_\ell/(1 + \text{Inf}_\ell)$.

Open problem. The empirical reversal of $\rho(\mu, |\text{supp}|)$ between $n = 4$ and $n = 5$ is not predicted by Theorem 3.8 in either direction — the theorem is a strict-Schur-convexity statement at fixed $I(f)$, which says μ *is monotone in concentration*, not that monotonicity transfers to minimum support at any n . The reversal we observe is therefore consistent with the theorem (which is a statement about μ alone, not about minimum support) but indicates that the empirical link between μ and minimum support is more subtle than a direct Schur-convexity transfer would suggest. We leave a precise theoretical characterisation as an open problem; a natural starting point is to ask whether μ becomes increasingly redundant at higher n relative to influence entropy, or whether the reversal is driven by the support distribution at $n = 5$ being too tight (mode 11 at 69.8% density) for the μ -induced ranking to surface signal.

8.5 Real-Valued Application (companion paper)

The BBT framework, while developed for Boolean functions, suggests a practical mechanism for quantizing real-valued weight matrices: per-coordinate spectral activation energy in the Walsh-Hadamard basis identifies which coordinates carry the most signal and therefore should be preserved most carefully under coarse quantization. A companion paper [2] reports an empirical validation of this idea on five pretrained decoder-only LLMs (SmolLM-135M/360M, Qwen2.5-0.5B/1.5B, TinyLlama-1.1B) at $W2A16$ (group size = 64), showing wikitext-2 perplexity reductions of 15–58% relative to vanilla `auto-round` [17], with the largest gains on models whose vanilla baselines are most degraded. Three architectural extensions handle non-trivial attention variants: a per-head PCA matrix- Γ replacement of q_norm/k_norm for Qwen3-style models, a $\text{SO}(2)$ -pair PCA that commutes with RoPE for non-norm models, and an MoE-aware `SCALETARGET` adapter for fused-expert layouts; a bit-width ablation at $W2$ vs. $W4$ shows the redistribution payoff scales with the per-channel quantization-noise budget, falling within the evaluation noise floor at $W4$. We separate this material into a companion paper because it relies on a real-valued spectral-energy proxy (per-coordinate WHT activation energy) that is *not* the Boolean influence of this paper; the connection is intuitive (per-coordinate Walsh-basis spectral mass) and the empirical perplexity improvements are substantial, but the transfer of the Schur-convexity argument (Theorem 3.8) from the Boolean to the real-valued setting is qualitative, not formal. We refer the reader to [2] for the full empirical results, ablations, implementation details, and the LLM-side limitations discussion.

8.6 What Is Proved vs. Conjectured

- **Proved here:** exact butterfly ℓ_p operator norms (Lemma 3.2, via Riesz–Thorin and duality); Jensen and Schur-convexity bounds for μ (Theorem 3.6, Theorem 3.8); rational/algebraic invariant properties (Propositions 5.1, 5.3); separation beyond total influence (Theorem 5.4).
- **Imported from companion work:** certified ternary Walsh-threshold representability for all Boolean functions through $n \leq 4$ (Theorem 7.1, cited from [1]).
- **Computed here:** exact MILP minimum-support certificates for all 65,536 Boolean functions at $n = 4$ (Section 7.1); enumeration of all 616,126 NPN-canonical Boolean functions at $n = 5$ (matching OEIS A000370); minimum-support MILP statistics on a 10,000-sample of those NPN-canonical reps (Section 8.4).
- **Empirical findings (not theorems):** marginal correlations of μ , influence entropy, max-influence, and cancellation diagnostics with minimum support are weak; conditional Spearman($\mu, |\text{supp}|$) at fixed $I = 1.75$ is $+0.571$ at $n = 4$ (the largest stratum), but reverses to ≈ -0.38 at $n = 5$ under both function-uniform and NPN-canonical sampling. Influence entropy continues to track support qualitatively at $n = 5$ where μ does not. These correlations are not theorems and do not establish that μ universally predicts support.
- **Open / conjectural:** ternary representability for $n \geq 5$ in full; polynomial-time synthesis under structural assumptions; theoretical explanation for the $n = 4$ to $n = 5$ reversal in $\rho(\mu, |\text{supp}|)$ (e.g. whether μ becomes increasingly redundant relative to influence entropy at higher n , or whether the reversal is driven by the support distribution being too tight at $n = 5$).

8.7 Algebraic Position of the Invariant

Because $\log_2 \mu(f)$ is rational rather than polynomial in the Fourier energy data, it sits structurally outside the class of low-degree polynomial spectral summaries (e.g. total influence, noise sensitivity, Fourier L^1 mass). We leave any relationship to algebraization or natural-proofs barriers [12] as an open question and make no claim that $\mu(f)$ escapes them; we present it only as a strictly finer invariant than total influence and as a practical diagnostic for spectral representation complexity.

8.8 Limitations

We make explicit the following limitations to forestall over-reading of the results.

- The contraction invariant $\mu(f)$ is a *function-dependent geometric diagnostic* derived from the influence-adaptive Walsh butterfly factorization. It is not a proof of ternary representability; the universality result through $n = 4$ is computational and certificate-based and does not imply universality for any $n \geq 5$.
- The cancellation index ρ is an empirical heuristic. We do not assert a global $\|w\|_1$ lower bound from layerwise non-cancellation; the pathwise step would require tracking per-coordinate butterfly subtree mass under additional non-cancellation hypotheses we do not formally establish. We therefore use ρ as a diagnostic correlate of support and repair difficulty (Remark 4.3), not as a proof component.

- The LLM quantization experiments use a real-valued spectral-energy proxy inspired by Boolean influence (per-coordinate WHT activation energy), *not* the Boolean influence itself. The connection between the two is intuitive (per-coordinate Walsh-basis spectral mass) and the empirical PPL improvements are real, but the transfer of the Schur-convexity prediction from the Boolean to the real-valued setting is qualitative.
- All LLM PPL numbers are single-seed and have an empirical ± 0.5 noise floor from calibration sampling; we treat differences smaller than this as no-effect data points.

8.9 Open Problems

1. Does ternary PTF universality extend to $n = 5$ and beyond?
2. Can the margin product $\mu(f)$ provide provable upper bounds on ternary PTF support size?
3. Is there a polynomial-time algorithm for ternary PTF synthesis when $I(f) = O(\log n)$?
4. How does the cancellation index ρ relate to circuit complexity?

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A Reproducibility of the Computational Universality Theorem

Theorem 7.1, imported from the companion synthesis manuscript [1], is a computational theorem verified by explicit certificate. For completeness, we describe the integer verification procedure used to check any supplied certificate so that reviewers can independently confirm it.

Certificate format. For each of the 2^{2^n} Boolean functions f on n variables, the certificate provides a ternary mask $w(f) \in \{-1, 0, +1\}^{2^n}$. The certificates for $n = 3$ (256 masks) and $n = 4$ (65,536 masks) originate in [1]; we redistribute them in the accompanying repository for this paper as supplementary material, alongside the new MILP minimum-support certificates produced in Section 7.1 and the 616,126 NPN-canonical $n = 5$ representatives produced in Section 8.4.

Verification procedure. For each function f and its certificate w :

1. Construct $H_n = H_1^{\otimes n}$ with $H_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ (integer matrix, no floating-point).
2. Compute $y = H_n w$ (integer matrix-vector product).

3. Verify $\text{sign}(y_i) = f_i$ for all $i \in [2^n]$.

At $n = 4$, this is $2^4 \times 2^4 = 256$ integer multiplications per function, with 65,536 functions total: approximately 16.8×10^6 operations, completing in under 1 second on any modern CPU.

Reproduction. The complete verification can be reproduced by running:

```
cd boolean_fourier
python3 -m bbt.exhaustive_validation --n 4 --repair
```

This regenerates all masks and verifies each one, producing the certificate file `results/exhaustive_n4_repaired.js`. Total runtime: approximately 5 seconds on a single CPU core.

B Clarkson Inequalities

Lemma 3.2 relies on the Clarkson inequalities [10]:

- For $1 \leq p \leq 2$ and $a, b \in \mathbb{R}$: $|a + b|^p + |a - b|^p \leq 2(|a|^p + |b|^p)$.
- For $2 \leq p \leq \infty$ and $a, b \in \mathbb{R}$: $|a + b|^p + |a - b|^p \geq 2(|a|^p + |b|^p)$.

Equality in the first case is attained when $ab = 0$ (one argument is zero), giving the ℓ_p operator norm maximizer at $(1, 0)$. Equality in the second case is attained at $|a| = |b|$, giving the maximizer at $(2^{-1/p}, 2^{-1/p})$.